

MASTER 1 GEOMATICS AND SPATIAL MODELING

HGABU36G01 - Applied Programming for GIS

Soft Mobility and Public Transit in Paris: Towards a Differentiated Accessibility of Intra-Parisian Space

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Link to access the notebook on Google Colab



https://colab.research.google.com/drive/12dIILAwTjImGbTliZ5ov2F_HIPy1MZocK?usp=sharing

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INTRODUCTION

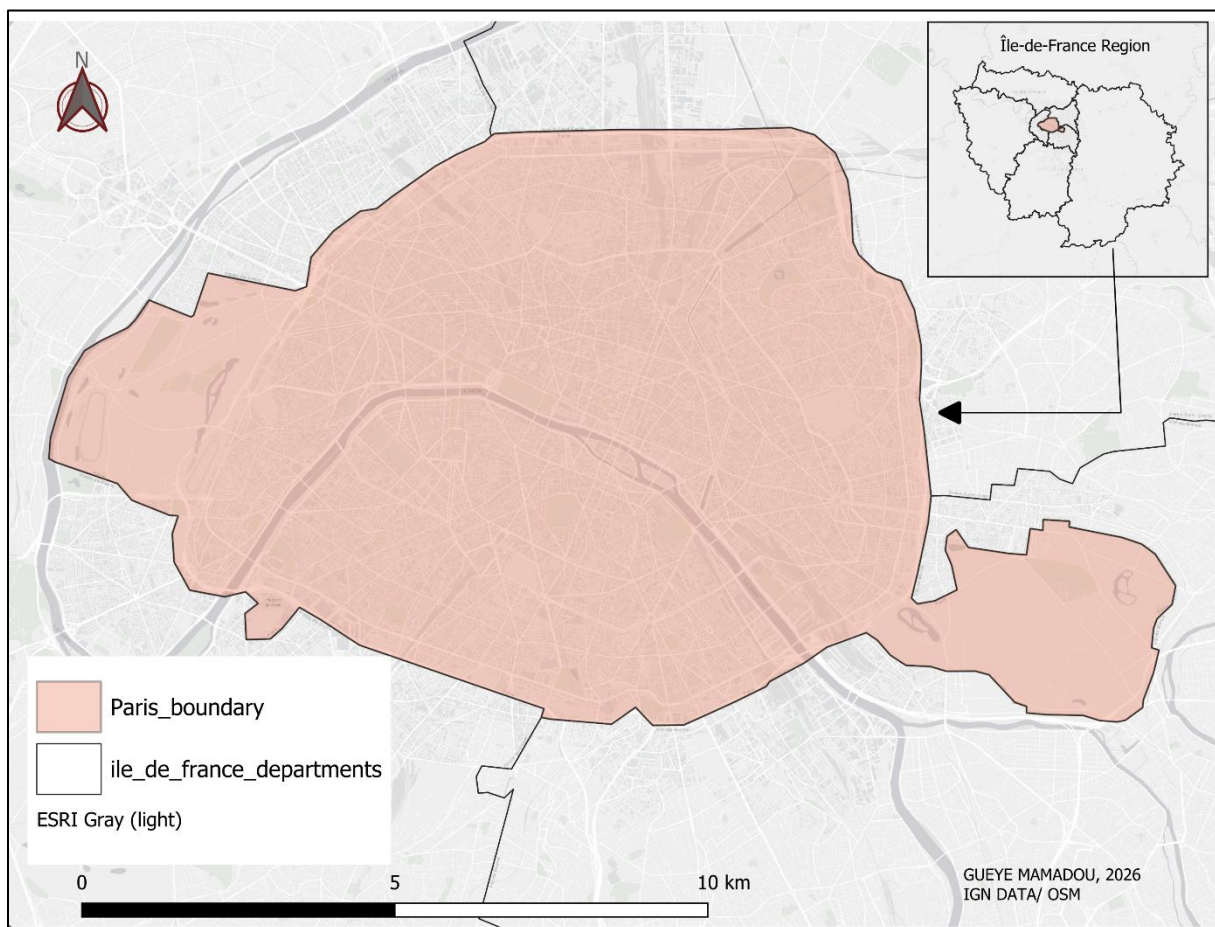
Paris operates one of the densest public transit networks in Europe, combining metro, tramway and bus lines across a territory of 105 km². Yet the mere existence of a rail network does not guarantee homogeneous accessibility for all residents. Since the 2010s, the rise of soft mobility bike-sharing schemes and structured cycling infrastructure has profoundly reshaped travel possibilities at the neighbourhood scale. The development of the 2026 cycling plan and the widespread deployment of Vélib' stations represent a major transformation of urban mobility supply, one that warrants analysis not in isolation, but in its articulation with existing public transit.

This complementarity lies at the heart of the present study. Drawing on a multimodal network model of Paris ; integrating GTFS data from Île-de-France Mobilités, the official cycling network and Vélib' stations ; we constructed a directed graph enabling the simulation of seven distinct mobility scenarios on a 200m × 200m grid covering the entirety of inner Paris. For each cell, the median travel time to all other cells is computed using Dijkstra's algorithm, yielding a granular and spatially explicit measure of accessibility under each modal configuration. This approach allows us not only to assess overall accessibility levels, but to isolate the individual contribution of each mode ; rail, bus, Vélib' ; and, crucially, to quantify the gains produced by their combination in areas where each mode, taken alone, remains insufficient.

The central question guiding this work is therefore: to what extent does the complementarity between soft mobility and public transit produce a differentiated accessibility across intra-Parisian space? More specifically, we examine whether Vélib', far from being a mere alternative mode, fulfils a corrective function with respect to spatial accessibility inequalities ; bridging the gaps left by the rail network and bus lines ; or whether, conversely, it amplifies pre-existing disparities by disproportionately benefiting already well-served areas. By mobilising INSEE sociodemographic data at the 200m grid level, this study cross-references accessibility and social vulnerability to assess whether the most fragile populations ; low-income households, elderly residents, social housing tenants ; genuinely benefit from this modal complementarity or remain structurally excluded from it.

Location and territorial context

Paris, the capital of France, is located at the heart of the Paris Basin, crossed by the Seine over approximately 13 kilometres. It covers an area of 105 km² and concentrates nearly 2.1 million inhabitants within its administrative boundaries, making it one of the densest metropolises in Europe with over 20,000 inhabitants per km². Paris is home to one of the densest and most integrated public transit systems in the world. The RATP network comprises 16 metro lines, 3 intra-urban tram lines and several hundred bus routes, complemented by the regional RER network. Since the 2010s, the city has undertaken a deep transformation of its mobility landscape in favour of soft modes: the 2026 cycling plan has extended the cycling network to over 1,000 kilometres of dedicated lanes, and the Vélib' bike-share system now counts more than 1,400 stations distributed across Paris and its surrounding municipalities. It is precisely this articulation between structuring public transit and soft mobility that forms the subject of the present study.



Map 1: Location of interest area (PARIS)

Part I : Tools and Working Environment

1.1 Spatial Processing Libraries ; GeoPandas, Shapely and Coordinate Reference System Management

1. GeoPandas

GeoPandas is the central spatial library of this project. It extends the Pandas DataFrame to handle geometric objects (points, lines, polygons) directly as a column, allowing all standard tabular operations ; filtering, merging, grouping ; to work on geographic data without any conversion step. In this notebook, GeoPandas loads every spatial layer (the 200m grid, the cycling network, Vélib' stations, IRIS boundaries, arrondissement polygons) and ensures they are all reprojected to Lambert-93 (EPSG:2154) before any distance or area calculation. The surface-weighted spatial join between the mobility grid and the INSEE Filosofi cells relies on `gpd.overlay`, which computes exact intersection geometries and is simply not available in plain Pandas. Compared to alternatives like ArcPy (proprietary) or R's `sf` (outside the Python ecosystem), GeoPandas offers the best trade-off: fully open-source, natively integrated with NumPy, Matplotlib and NetworkX, and capable of reading all formats used here (GeoPackage, GeoJSON, Shapefile).

Example:

```
import geopandas as gpd
maillles = gpd.read_file("resultats200m_v4_paris.gpkg").to_crs(2154)
districts = gpd.read_file(url_districts).to_crs(2154)
intersection = gpd.overlay(maillles, carreaux, how='intersection')
intersection['ratio'] = intersection.geometry.area / intersection['cell_area']
```

2. Shapely

Shapely is the geometry engine that GeoPandas relies on under the hood, but it is also used directly in this project wherever individual geometric operations need to be performed on specific objects rather than on an entire column. Shapely provides the full set of computational geometry primitives ; centroid, buffer, intersection, distance, within ; with a clean and readable Python API. In the graph construction phase, Shapely is used to extract cell centroids and to identify the nearest cycling node or transit stop to each centroid by computing point-to-point distances. It is also used to build the 400-metre

walking catchment buffers around Vélib' stations that define the intermodal transfer radius. Without Shapely, these operations would require raw coordinate arithmetic or an external GIS tool. No comparable pure-Python alternative exists: PyGEOS (now merged into Shapely 2.0) was its only direct competitor and has been fully absorbed into it.

Example:

```
from shapely.geometry import Point
maillles['centroid'] = maillles.geometry.centroid
stations['buffer_400m'] = stations.geometry.buffer(400)
stop_point = Point(2.3522, 48.8566)
print(stations['buffer_400m'].contains(stop_point))
```

3. Pyproj

Pyproj handles all coordinate reference system transformations at the lower level. While GeoPandas calls Pyproj internally when `.to_crs()` is invoked, Pyproj is also used directly when a precise geodesic distance (in metres along the Earth's surface) needs to be computed between two geographic coordinates, or when a custom CRS transformation must be applied to individual points rather than to a full GeoDataFrame. In this project, it is particularly useful during the graph construction phase when transit stop coordinates from the GTFS feed (provided in WGS84, EPSG:4326) need to be converted to metric Lambert-93 coordinates before computing access distances. The alternative, `osgeo.osr` from GDAL, provides the same functionality but with a significantly more verbose and less Pythonic API. Pyproj is the standard choice across the Python geospatial ecosystem precisely because of its simplicity and its direct wrapping of the PROJ library.

```
from pyproj import Transformer
transformer = Transformer.from_crs("EPSG:4326", "EPSG:2154",
always_xy=True)
lon, lat = 2.3522, 48.8566
x, y = transformer.transform(lon, lat)
print(f"Lambert-93: x={x:.1f}, y={y:.1f}")
stops['x'], stops['y'] = transformer.transform(
    stops['stop_lon'].values,
    stops['stop_lat'].values)
```


)

1.2 Network Modelling ; NetworkX, Directed Graphs and the Dijkstra Algorithm

1. NetworkX

NetworkX is the graph library that implements the multimodal accessibility model at the core of this study. The entire transport network ; metro, tram, bus, cycling arcs, Vélib' access edges, intermodal transfers, and pedestrian grid connections ; is encoded as a single weighted DiGraph (directed graph), where each node is a stop, station or grid cell, and each edge weight represents a travel time in seconds. The `single_source_dijkstra_path_length` function then computes, from each origin cell, the shortest travel time to every reachable destination. The median of those times is the accessibility indicator. NetworkX is chosen over alternatives like `igraph` or `graph-tool` because it integrates seamlessly with Pandas and GeoPandas for building the graph from spatial DataFrames, and because its Python-native objects make debugging and inspection straightforward. For the scale of inner Paris at 200m resolution, NetworkX's performance is entirely adequate; `graph-tool`'s C++ backend would only become necessary at national scale.

Example:

```
import networkx as nx
G = nx.DiGraph()
G.add_edge("stop_A", "stop_B", weight=180) # 3 minutes
G.add_edge("node_12", "node_45", weight=94) # distance/speed
times = nx.single_source_dijkstra_path_length(G, source="orig_42",
cutoff=7200)
median_time = sorted(times.values())[len(times)//2] / 60 # convert to
minutes
```

2. OSMnx

OSMnx is used to retrieve and process the street network from OpenStreetMap, which serves as the walking layer for pedestrian access arcs in the graph. It downloads the drivable or walkable street graph for a given area, simplifies it, and returns it directly as a

NetworkX MultiDiGraph, meaning the resulting graph can be merged with the transit and cycling graphs without any conversion. In this project, OSMnx provides the walking network used to estimate access times between grid cell centroids and the nearest transit stop or Vélib' station. It also enables the computation of realistic walking distances along the street network rather than straight-line Euclidean distances, which is important for accuracy in dense urban environments like Paris where buildings block direct paths. The alternative would be to use straight-line distance with a detour factor, which is a cruder approximation. No other library provides the same combination of OSM download, graph simplification, and direct NetworkX output.

Example:

```
import osmnx as ox
G_walk = ox.graph_from_place("Paris, France", network_type="walk")
nearest_node = ox.nearest_nodes(G_walk, X=2.3522, Y=48.8566)
walking_times = nx.single_source_dijkstra_path_length(
    G_walk, source=nearest_node, cutoff=900
)
```

3. Pandas

While not a graph library, Pandas is indispensable in the network construction phase because all source data ; GTFS tables (stops, stop_times, trips, routes), Vélib' station attributes, and grid cell identifiers ; arrive as tabular CSV or GeoPackage files. Pandas handles the parsing, filtering, merging and restructuring of these tables before any graph edge can be created. For example, the GTFS stop_times table is joined with trips and routes to recover the route type for each inter-stop arc, which determines the travel time penalty applied. The iterrows or vectorised operations then feed edge lists directly into G.add_edges_from(). Pandas is also used throughout the accessibility computation phase to store, aggregate and export the Dijkstra results per cell before passing them to GeoPandas for mapping. Polars would be faster for very large GTFS feeds, but Pandas's tighter integration with GeoPandas and NetworkX makes it the practical choice at this scale.

Example:

```

import pandas as pd
stop_times = pd.read_csv("stop_times.txt")
trips      = pd.read_csv("trips.txt")
routes     = pd.read_csv("routes.txt")

merged = (stop_times
          .merge(trips[['trip_id','route_id']], on='trip_id')
          .merge(routes[['route_id','route_type']], on='route_id')
          .sort_values(['trip_id','stop_sequence'])
          )
edges = []
for _, grp in merged.groupby('trip_id'):
    for i in range(len(grp) - 1):
        o, d = grp.iloc[i], grp.iloc[i+1]
        edges.append((o['stop_id'], d['stop_id'], {'weight': 60}))
G.add_edges_from(edges)

```

1.3 Visualisation Libraries ; Folium, Matplotlib and Plotly

1. Folium

Folium is the library used for all interactive maps in this notebook. It wraps Leaflet.js and generates self-contained HTML files that embed the map, the data, the colour scale, and the layer controls without requiring any JavaScript to be written. In this project, Folium renders the reference accessibility map (m1), the mobility desert map (m2), the modal contribution map (m3), the poverty overlay (m_pov), the K-Means cluster map (m_clusters), and the arrondissement-level choropleth (m_arr). Its key advantages over static mapping are the ability to add tooltips on each grid cell (showing travel time, population, and social indicators on hover), a layer control that allows the user to toggle between modal contribution layers on a shared colour scale, and proportional circle markers that communicate population exposure in the desert zones. Compared to ipyleaflet, which offers similar interactivity but is tightly bound to the Jupyter kernel,

Folium produces standalone HTML files that can be shared, embedded in a report, or published without a running Python environment.

Example:

```
import folium
import branca.colormap as cm
cmap = cm.LinearColormap(['#2166ac','#f7f7f7','#d73027'],
                          vmin=15, vmax=40, caption="Travel time (min)")

m = folium.Map(location=[48.856, 2.352], zoom_start=12,
tiles="cartodbpositron")
folium.GeoJson(
    mailles_4326[['geometry','ref_total','population']],
    style_function=lambda x: {
        'fillColor':  cmap(x['properties']['ref_total']),
        'color':      'none',
        'fillOpacity': 0.75
    },
    tooltip=folium.GeoJsonTooltip(
        fields=['ref_total','population'],
        aliases=['Travel time (min):', 'Population:']
    )
).add_to(m)
cmap.add_to(m)
m.save("map_reference.html")
```

2. Matplotlib

Matplotlib handles all static, publication-quality figures in this study. Its fine-grained control over every visual element ; axis labels, tick formatters, annotation positions, shaded regions, line styles, multi-panel layouts via GridSpec ; makes it the right tool when a figure needs to be precise, reproducible and exportable at high resolution (300 dpi PNG or vector SVG/PDF for academic output). In this notebook, Matplotlib produces the population-weighted histogram of travel times (with the weighted median and mean

overlaid as reference lines and the critical zone shaded), and the Lorenz curves (with fill-between shading for each scenario and axis-level annotation). These are figures where the exact placement of text, lines and colour zones matters, and where Matplotlib's imperative API gives complete control. Seaborn would simplify some of the statistical plots, but it abstracts away the axis-level control needed here; Plotly would make them interactive but less suitable for print output.

Example:

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(10, 5))
ax.hist(data['ref_total'], bins=30, weights=data['population'],
        color='#4393c3', edgecolor='white', alpha=0.85)
ax.axvline(weighted_median, color='#d73027', linestyle='--', linewidth=2,
          label=f'Weighted median: {weighted_median:.1f} min')
ax.axvspan(threshold, data['ref_total'].max(),
          alpha=0.12, color='#d73027', label='Critical zone')
ax.set_xlabel("Median travel time (min)")
ax.set_ylabel("Resident population")
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
ax.legend()
plt.tight_layout()
plt.savefig("histogram.png", dpi=300)
```

3. Plotly

Plotly produces all interactive statistical charts in this notebook. Its key advantage over Matplotlib is that every chart element ; bars, scatter points, histogram bins, Lorenz curve traces ; carries a hover label, so the reader can inspect exact values without cluttering the figure with annotations. In this study, Plotly builds the multi-panel synthesis dashboard (6 subplots combining bar charts, a summary table, a histogram and Lorenz curves in a single figure), the scenario boxplot for travel time comparison, the correlation scatter plots with population-weighted regression lines, and the arrondissement ranking bar chart. The `make_subplots` function with mixed spec types (bar, table, histogram, scatter in the same figure) is a capability that Matplotlib can replicate but with significantly more

boilerplate. Plotly figures also export to clean PNG via `fig.write_image()` and render natively in Jupyter without any backend configuration.

Example:

```
import plotly.graph_objects as go
from plotly.subplots import make_subplots
fig = make_subplots(rows=1, cols=2,
                    subplot_titles=("Mean travel time by scenario",
                                    "Gini index by scenario"))

fig.add_trace(go.Bar(x=labels, y=means, marker_color=colours,
                    text=[f'{v} min' for v in means],
                    textposition='outside'), row=1, col=1)
fig.add_trace(go.Bar(x=labels, y=ginis, marker_color=colours,
                    text=[f'{v:.3f}' for v in ginis],
                    textposition='outside'), row=1, col=2)
fig.update_layout(title="Accessibility indicators by scenario",
                  template='plotly_white', height=450, showlegend=False)
fig.show()
```

Part II : Data and Methodology

2.1 Data Sources

GTFS data from Île-de-France Mobilités provides the backbone of the public transit network: stops, routes, trips and stop_times tables are parsed to extract inter-stop travel times by mode (metro, tram, bus), distinguishing mode types by the GTFS route_type field (tram = 0, metro = 1, bus = 3). The Paris 2026 Cycling Plan (GeoJSON) supplies the cycling network geometry, encoded as bidirectional arcs with travel time derived from segment length divided by an assumed cycling speed of 15 km/h. Vélib' station data (GeoJSON) provides the locations of bike-share docking stations, which function as intermodal transfer nodes in the graph. INSEE Filosofi 200m supplies population counts and social indicators at a spatial resolution that matches the mobility grid exactly, enabling direct spatial joins without resampling.

2.2 Construction of the Multimodal Graph

The multimodal directed graph is built in NetworkX by stacking five arc types onto a single DiGraph. Transit arcs connect consecutive stops along each route, weighted by scheduled travel time. Cycling arcs connect adjacent nodes of the cycling network, bidirectional and weighted by distance over speed. Vélib' access arcs connect stations to the nearest cycling network node, adding a 30-second penalty to represent the unlocking and locking process. Intermodal transfer arcs link Vélib' stations to transit stops within a 400-metre walking catchment, adding a 120-second boarding penalty. Grid access arcs connect each 200m cell centroid to its nearest transit stops, cycling nodes, and Vélib' stations via walking time. This layered construction allows any combination of modes to be activated or deactivated by simply filtering the arc types included in the graph before running Dijkstra, which is the mechanism behind the seven comparative scenarios.

2.3 The 200m Grid as the Unit of Analysis

A 200m × 200m regular grid in Lambert-93 covers the full extent of inner Paris (approximately 105 km²). Each cell has one origin node and one destination node. For each origin, `single_source_dijkstra_path_length` returns the minimum travel time to all reachable destinations; unreachable cells are penalised at 120 minutes. The accessibility indicator for each cell is the median of all destination travel times, which is more robust than the mean to the penalised unreachable cells. This produces a continuous surface of accessibility scores that can be mapped, aggregated by arrondissement, crossed with sociodemographic data, and compared across the seven scenarios.

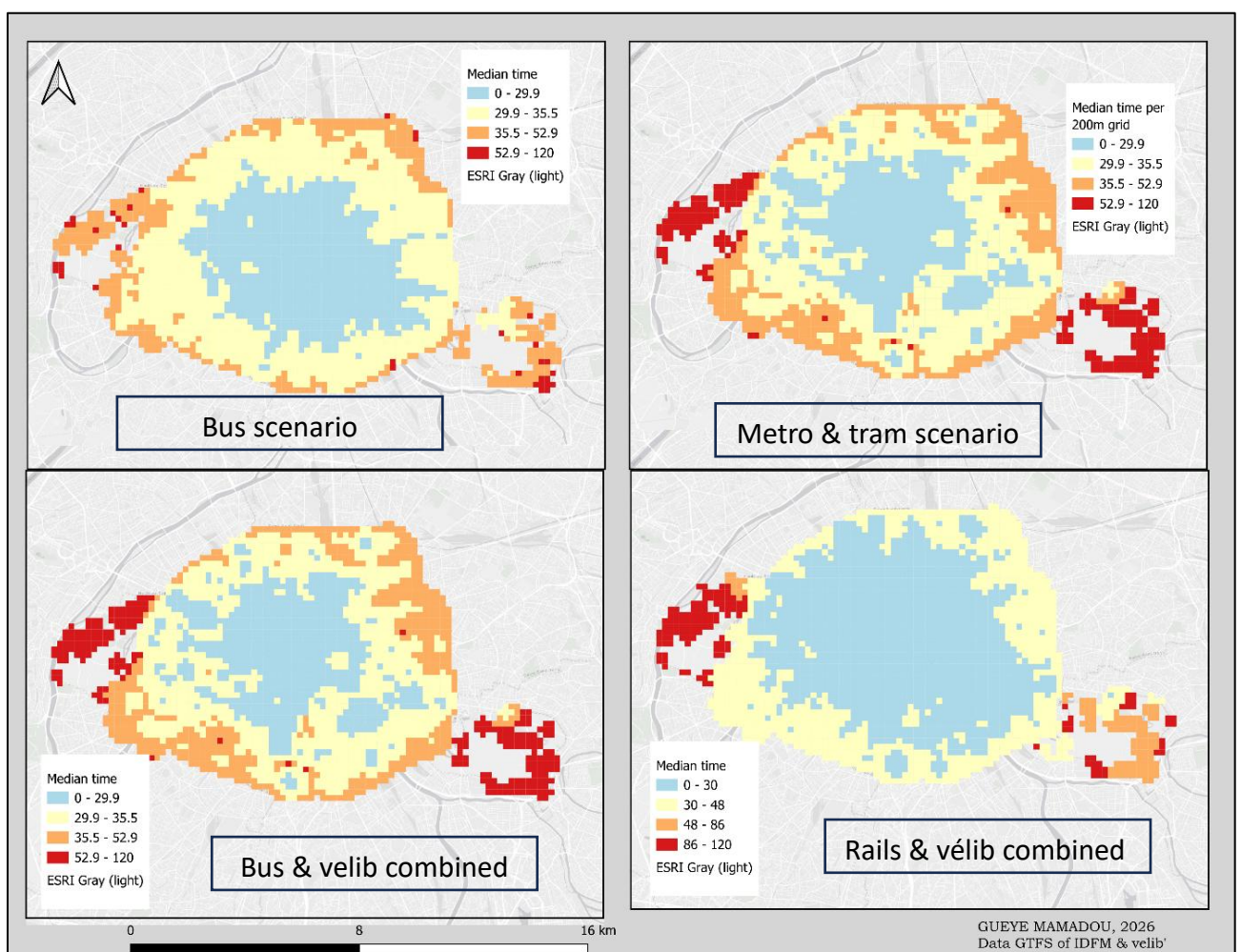
2.4 Integration of Social Data

The spatial join between the mobility grid and the INSEE Filosofi cells uses a surface-weighted proportional allocation: for each fragment produced by `gpd.overlay`, the fragment's area as a share of the parent INSEE cell's area determines the proportion of that cell's demographic count attributed to the overlapping mobility grid cell. This method is more accurate than a simple centroid join, particularly along cell boundaries. Five indicators are retained: total population (`ind`), low-income households (`men_pauv`), social housing units (`log_soc`), population aged 65–79 (`ind_65_79`), and population aged 80 and over (`ind_80p`). These serve as population weights throughout the analysis and as vulnerability indicators in Part III.

Part III : Results

3.1 Reference Accessibility in Paris

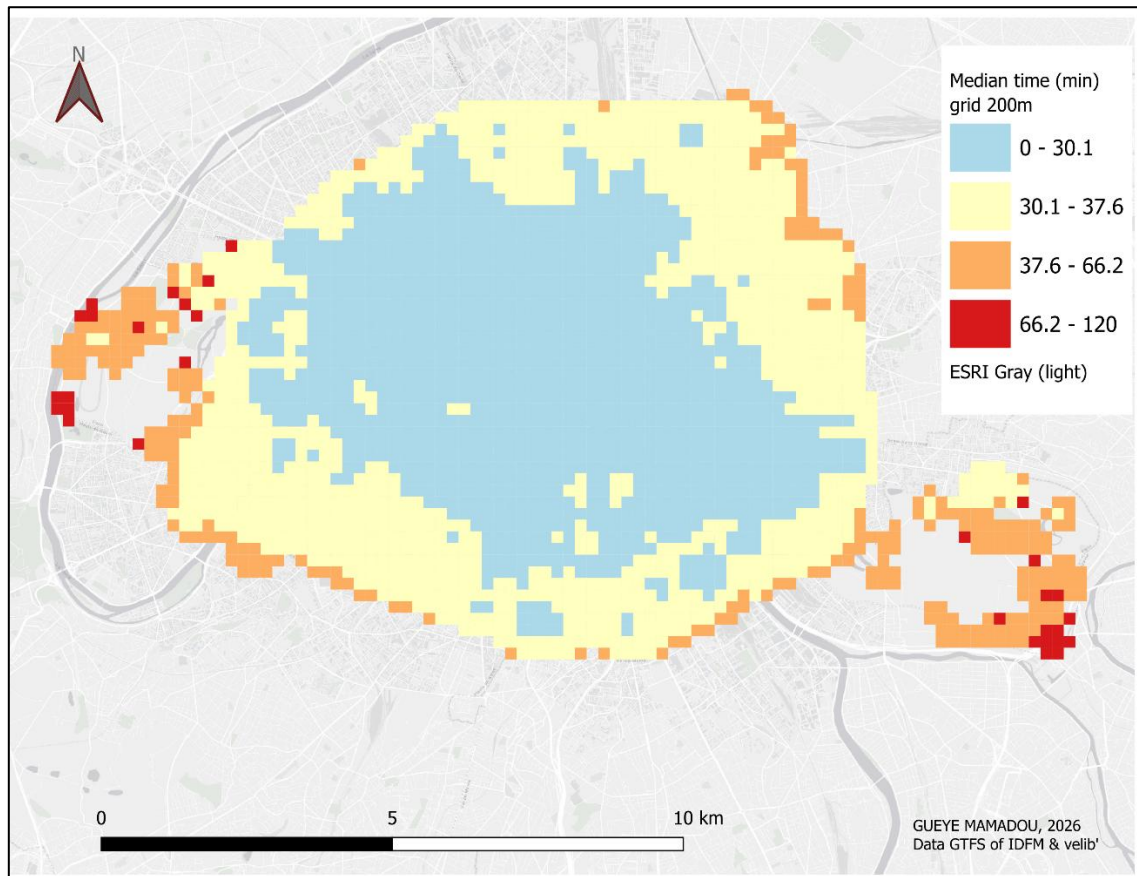
This composite map presents four simulated mobility scenarios on a 200m grid covering inner Paris. Median travel times range from under 30 minutes at the city's core to over 120 minutes at the periphery or in poorly served areas. The reference scenario (all modes combined) produces the most homogeneous pattern, with a well-connected centre and progressively degraded peripheral fringes.



Map 2: Comparison of Accessibility Scenarios by Mode of Transport per 200m Grid Cell

Comparing the four panels reveals the relative importance of each mode. Removing the rail network (bus + Vélib' scenario) strongly degrades accessibility in the dense core, while removing the bus affects peripheral areas where the metro is absent more severely. The

16th arrondissement (Bois de Boulogne) and the 12th arrondissement annex (Bois de Vincennes) consistently emerge as critical zones across all scenarios, regardless of the mode considered.

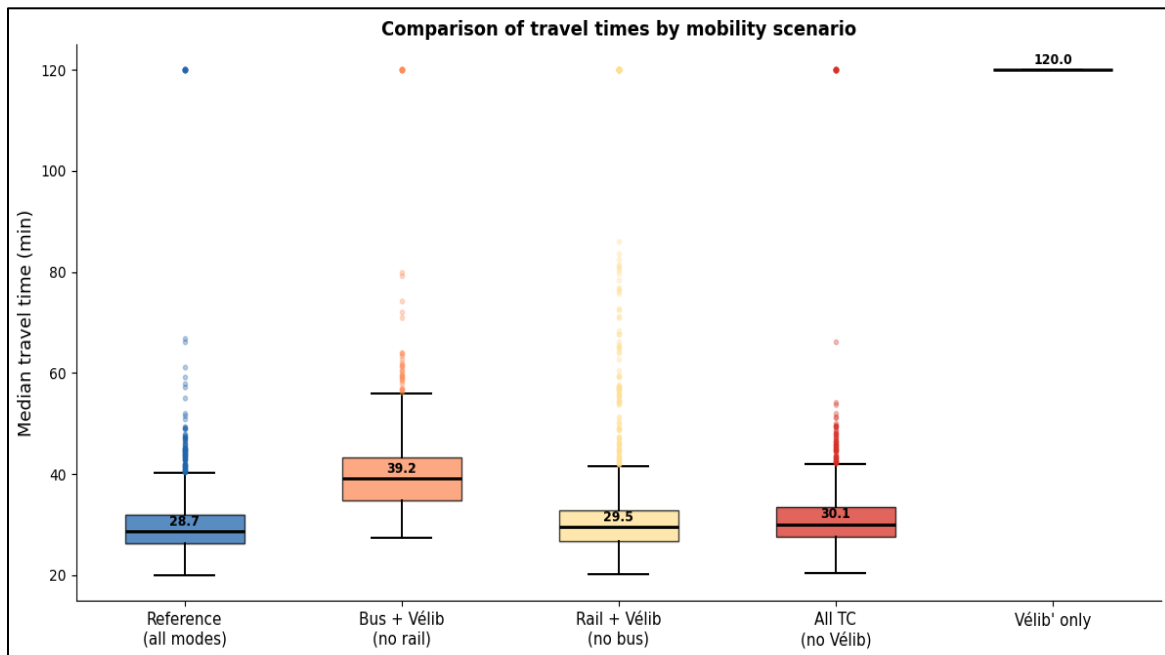


Map 3: Scenario Public Transport Accessibility Without Vélib'

This map shows median travel times under the scenario where all public transit (metro, tram, bus) is available but without the Vélib' bike-share system. The historic centre (1st–4th arrondissements) displays travel times under 30 minutes thanks to the exceptional density of the metro network. A pale yellow ring (30–37 minutes) surrounds this core, corresponding to arrondissements well served by several metro and tram lines.

Orange and red zones (37–120 minutes) concentrate at the periphery, especially in the 16th (Bois de Boulogne) and the 12th arrondissement annex (Bois de Vincennes), which form low-accessibility peninsulas in the absence of Vélib'. These large green spaces, sparsely served by PT stops, reveal the structural limits of the network when not supplemented by soft mobility. This scenario serves as the baseline for measuring the marginal contribution of Vélib'.

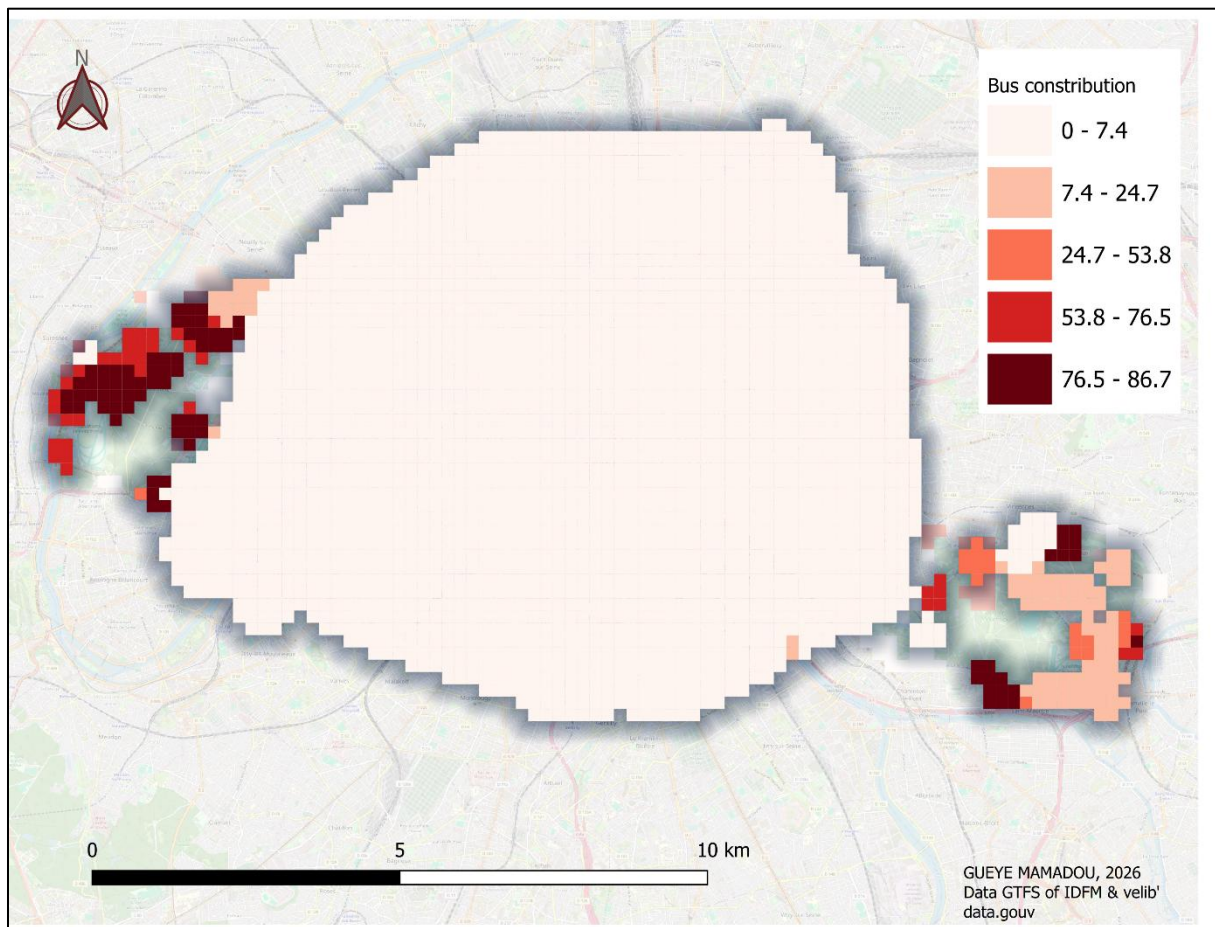
Figure 1: Comparison of Travel Times by Mobility Scenario (Box Plots)



This box plot chart compares the distribution of median travel times for five scenarios. The reference scenario (all modes, median 28.7 min) has the most compact box, indicating a tight and equitable distribution. The Bus + Vélib' scenario (median 39.2 min) shows the highest and widest box, confirming that removing the rail network not only degrades the average level but also increases spatial inequalities. The Rail + Vélib' scenario (29.5 min) is very close to the reference, suggesting that the bus contributes little to overall accessibility where the metro is present. The Vélib'-only scenario (median 120 min) is a limit case: all cells receive the maximum penalty (120 min = unreachable destination), reflecting the inability of bike-share alone to ensure complete intra-Parisian travel at city scale. The outliers present in each scenario correspond to mobility desert cells (Bois de Boulogne, Bois de Vincennes) that display extreme times regardless of the modal combination. The low dispersion of the reference scenario confirms the redistributive efficiency of multimodality.

3.2 The Contribution of Each Mode

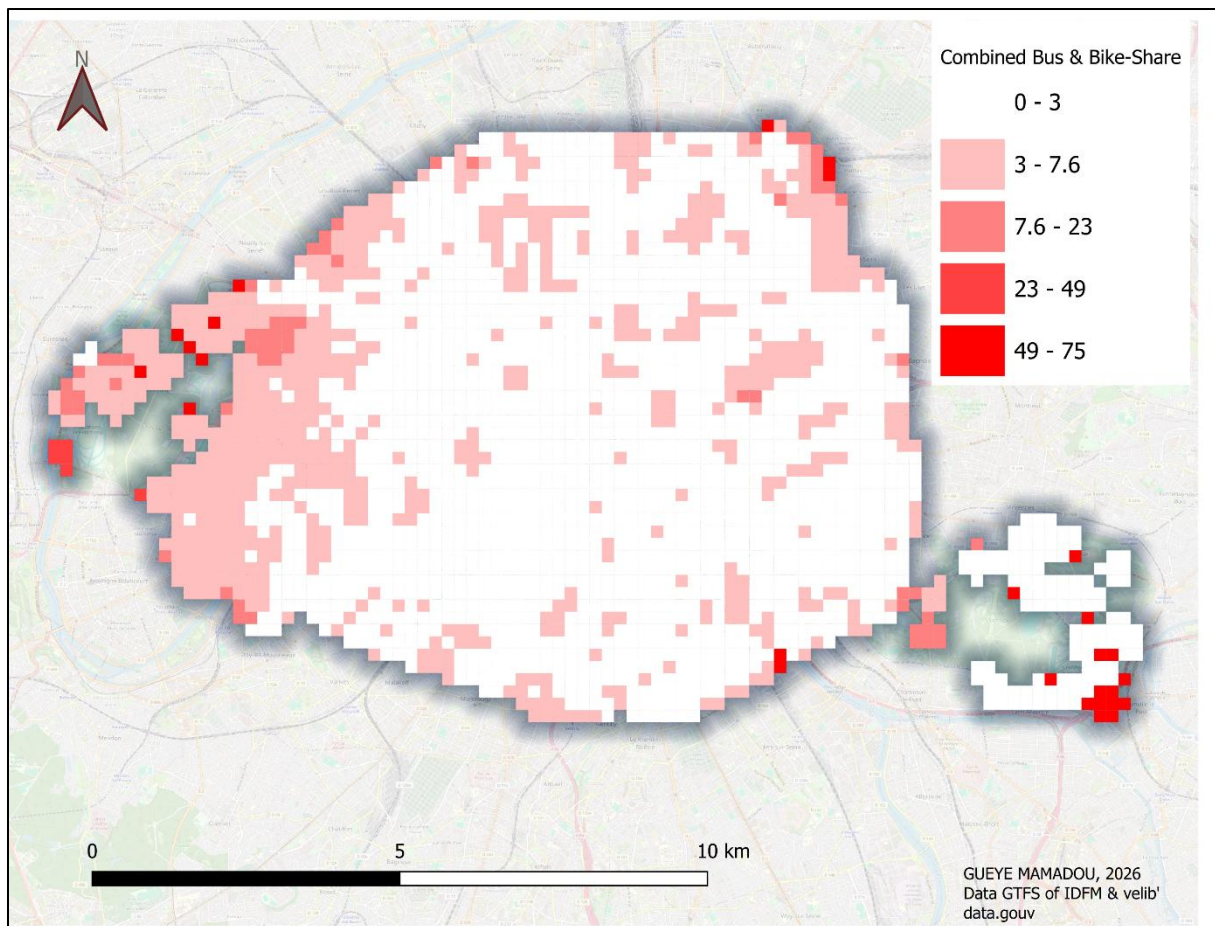
Bus Network Contribution / Minutes Saved Thanks to Bus Lines



Map 4: Bus Network Contribution & Minutes Saved Thanks to Bus Lines

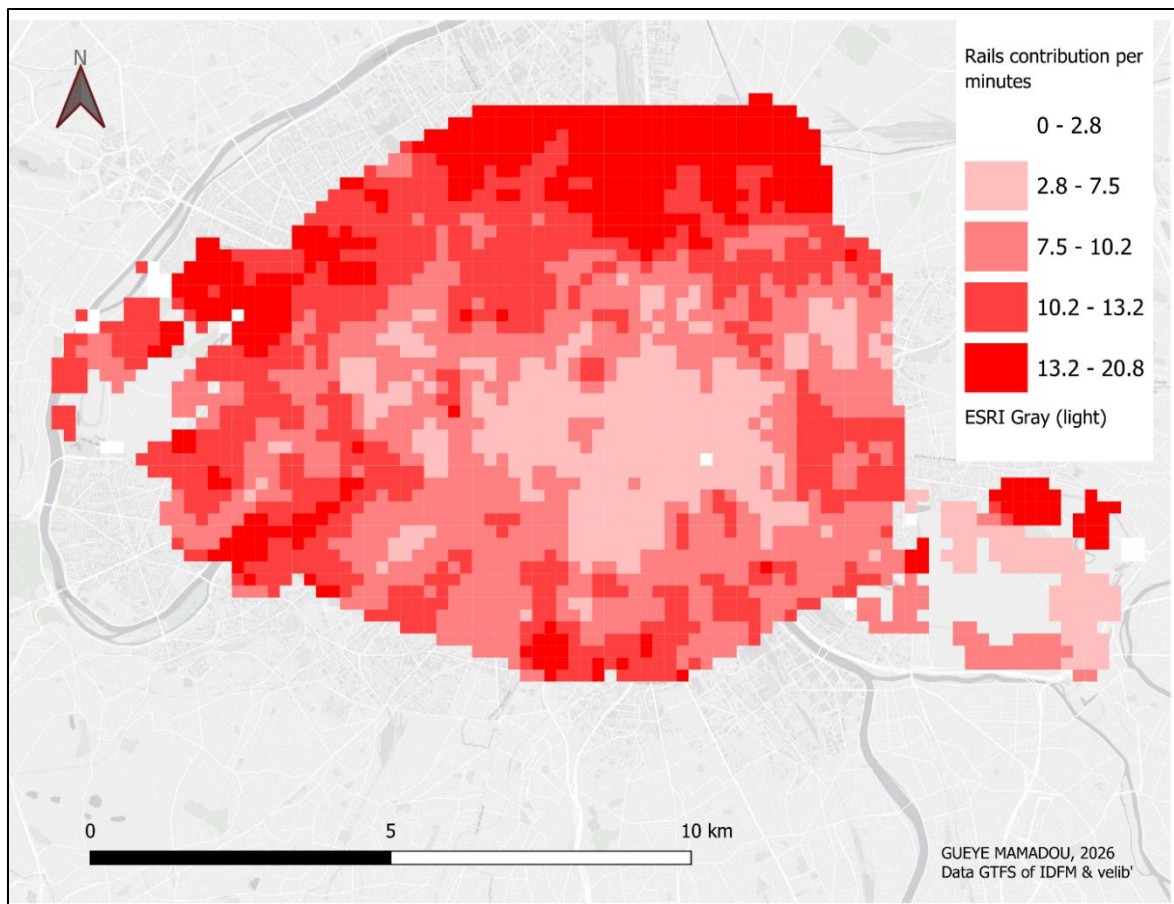
This map reveals a strongly contrasted geography of the bus contribution. The vast majority of inner Paris cells shows zero or very low contribution (white to very pale pink shading, 0–7.4 minutes), meaning that where the metro is available, the bus provides only marginal gains. The bus plays a fine-grained complementarity role rather than a substitution one. This result underlines the modal hierarchy in Paris: rail is the structuring mode, and the bus is the adjustment mode.

Two exceptional zones stand out clearly. In the west, the 16th arrondissement (Bois de Boulogne sector), the bus contributes between 53 and 86 minutes of gain, making it a near-essential mode for areas not covered by the metro. The 12th arrondissement annex (Bois de Vincennes) presents the same configuration. These two enclaves confirm that the bus fulfils a last-resort service function in forested sectors, where it is the only available public transit mode, making it a critical infrastructure for territorial equity.



Map 5: Combined Bus & Vélib' Contribution (Accessibility Gain time from the Two Modes Combined)

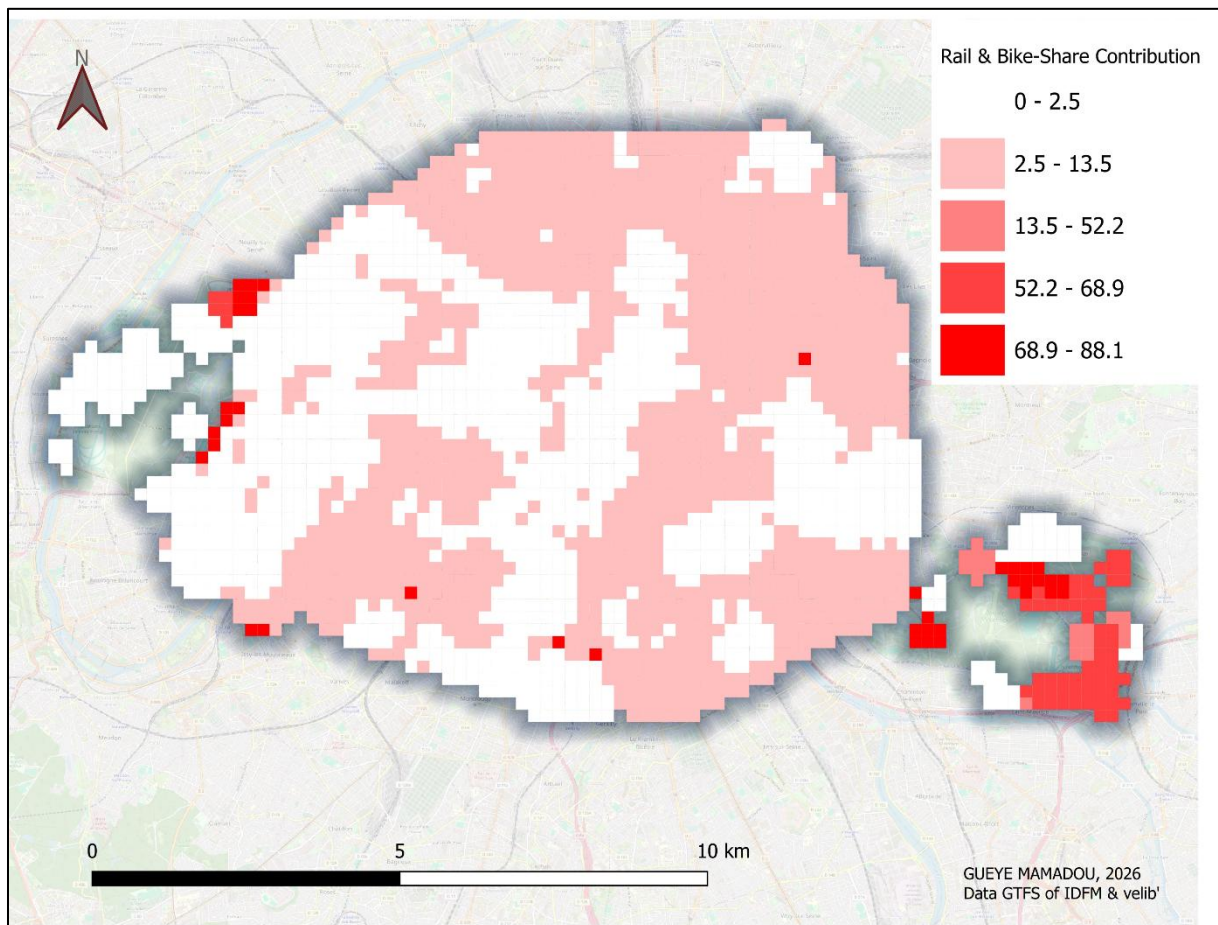
This map represents the accessibility gain linked to the complementarity between bus and Vélib'. the additional benefit brought by Vélib' when used in combination with the bus. Pale pink covers most of Paris, indicating a generalised but moderate gain (3–23 minutes). This homogeneous spatial spread suggests that Vélib' meaningfully improves bus accessibility across the whole city. High-gain zones (23–75 minutes, orange to bright red) concentrate in the peripheral western arrondissements (16th) and the 12th annex, where the metro network is absent and where the bus + bike combination becomes the only high-performing alternative. These zones coincide with the mobility deserts identified in the previous maps. Vélib' thus plays an amplifying role for bus accessibility: it enables faster transfers, reduces waiting times, and serves capillary areas not covered by bus lines.



Map 6 : Rail Network Contribution and minutes Saved Thanks to Metro and Tram

This map measures the rail network contribution (metro + tram) by computing the travel time difference between the bus + Vélib' scenario and the full reference scenario. Intense red covers nearly all of inner Paris, indicating that the rail network saves between 2.8 and 20.8 minutes across almost the entire territory. This benefit is particularly pronounced in the north-eastern arrondissements (10th, 18th, 19th, 20th) and the south (13th, 14th), where metro lines and trams form the backbone of travel.

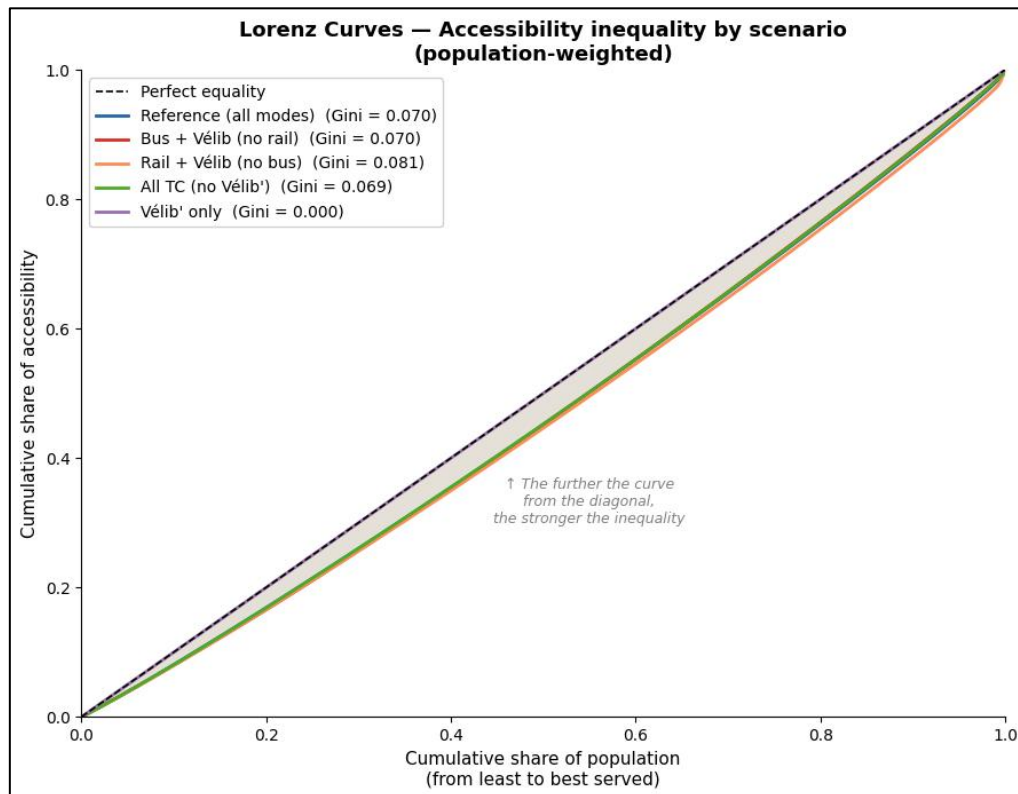
The centre of Paris (1st–4th) paradoxically displays lighter shading (2.8–7.5 minutes), not because rail is less useful there, but because the multimodal density of the centre already makes travel very fast in all scenarios. By contrast, peripheral areas benefit most from rail presence, as without it they would face substantially higher travel times. The map's homogeneity confirms that the Paris metro plays a universal structuring role that is not geographically differentiated.



Map 7: Combined Rail + Vélib' Contribution and Accessibility gain from Rail and Bike-Share Combined (minutes)

This result illustrates the value of first and last kilometres: Vélib' reduces access time to stations and extends their effective catchment area. High-gain zones (52–88 minutes, bright red) concentrate in two sectors: the 16th arrondissement and the 12th annex ; precisely the same enclaves identified as mobility deserts. In these sectors, the rail + Vélib' combination represents the only way to substantially reduce travel times, as rail network density is too low to cover these spaces. This result advocates for a priority deployment policy for Vélib' stations in peripheral zones with low rail density.

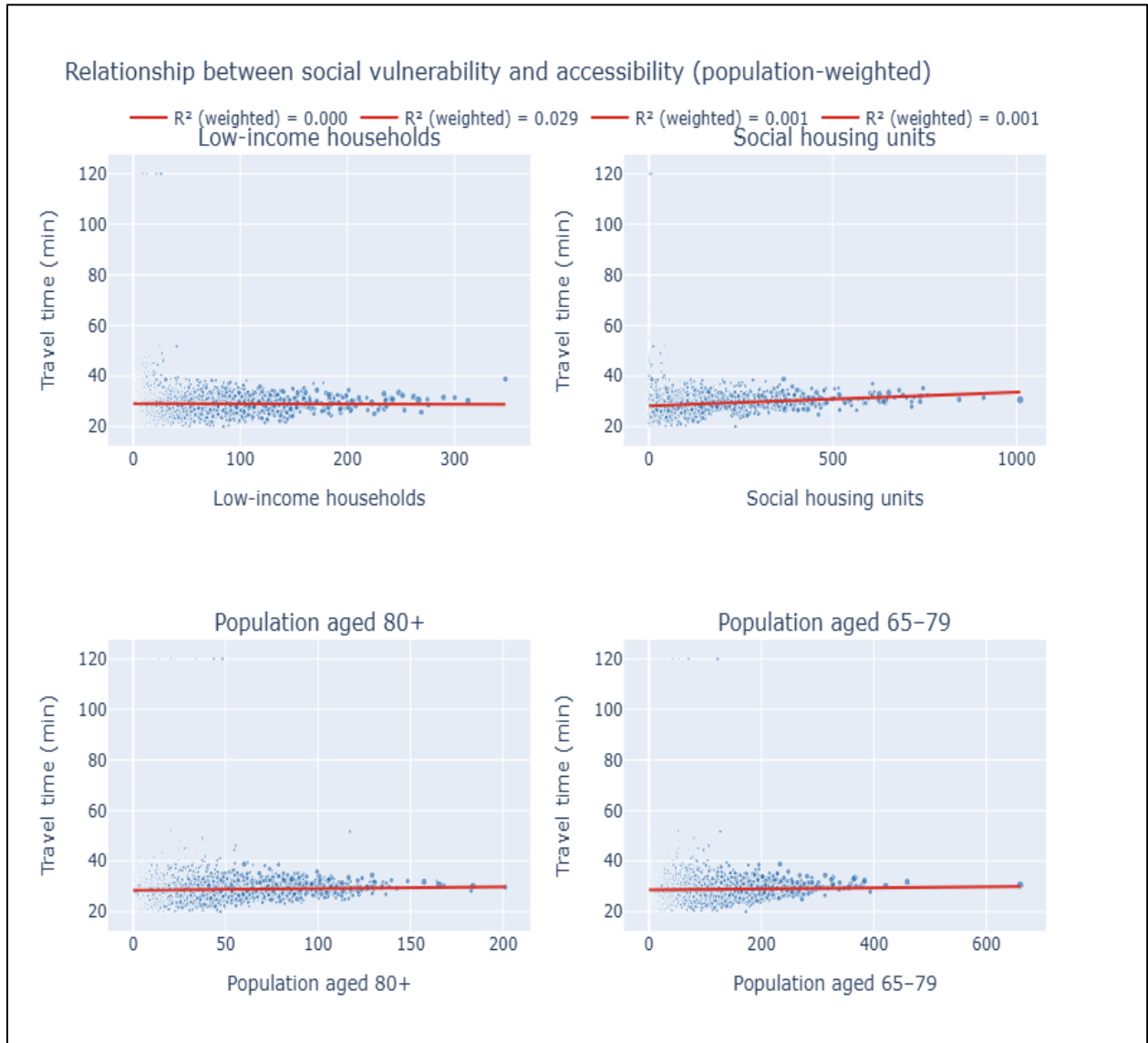
Figure 2 : Accessibility Inequality by Scenario (Population-Weighted)



The Lorenz curves measure the inequality of accessibility distribution among residents. The diagonal represents perfect equality. All scenarios display curves very close to the diagonal, with low Gini indices (0.069–0.081), confirming that the Parisian network distributes accessibility relatively equitably. The Rail + Vélib' curve (Gini = 0.081) deviates slightly more from the diagonal, indicating that removing the bus marginally increases inequality, likely by penalising peripheral zones already at risk.

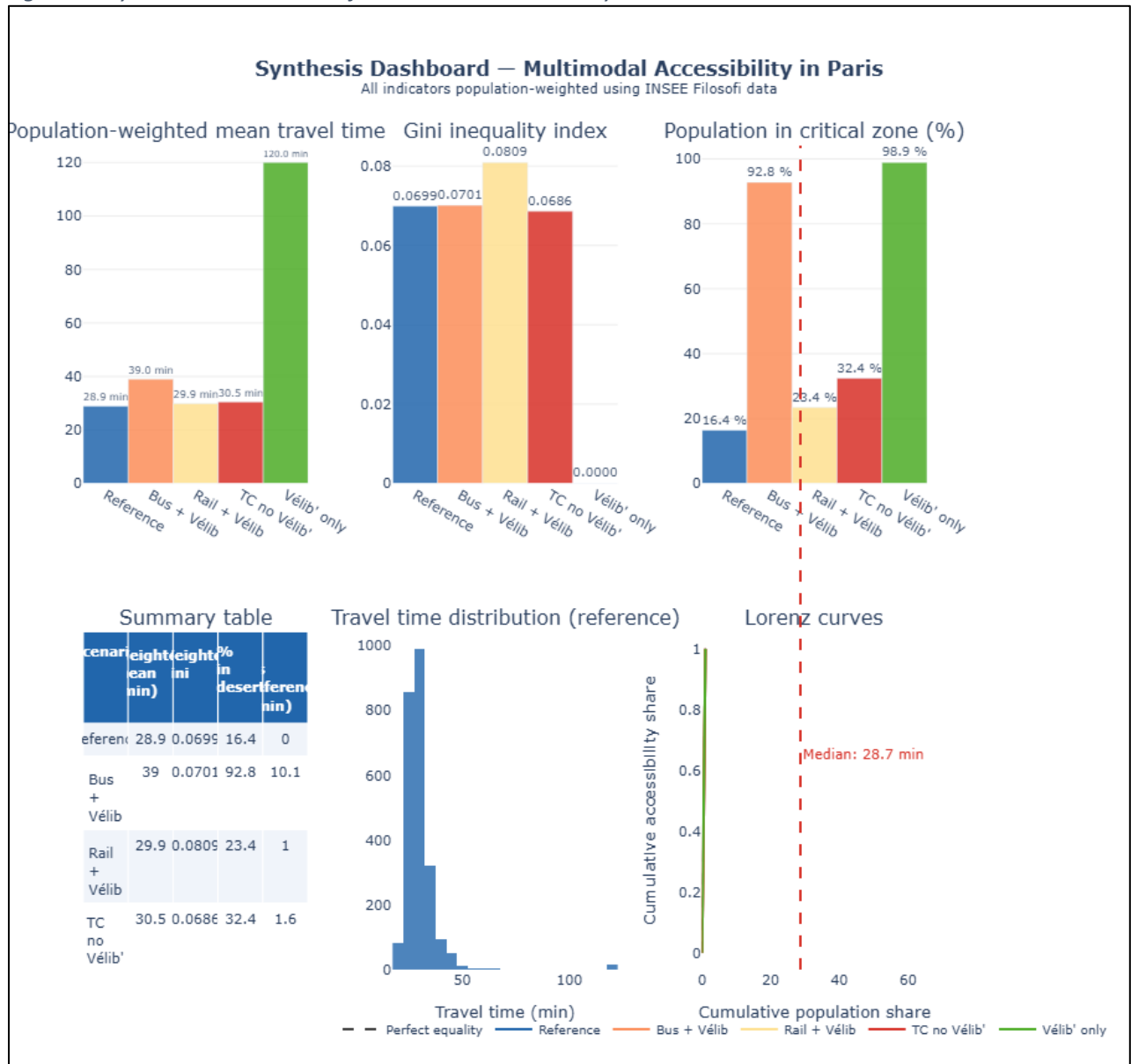
The most counter-intuitive result is that of the Vélib'-only scenario (Gini = 0.000, curve overlapping the diagonal): perfect equality here is achieved because all cells are equally inaccessible (120 min time), not because they all benefit from equivalent service. This illustrates the paradox of equality in deprivation rather than in access. The proximity of the reference and TC no Vélib' curves (Gini 0.070 and 0.069 respectively) indicates that Vélib' contributes very marginally to equity, its impact remaining localised to specific zones.

Figure 3 : Relationship Between Social Vulnerability and Accessibility (Population-Weighted Scatter Plots)



These four scatter plots cross social vulnerability indicators (low-income households, social housing units, 80+ population, 65-79 population) with each cell's median travel time, weighted by resident population. All four R^2 coefficients are extremely low (0.000 to 0.029), meaning there is virtually no linear relationship between the concentration of vulnerable populations and accessibility level at the 200m cell scale. This finding is analytically fundamental: it refutes the hypothesis of systematic accessibility discrimination against precarious populations in Paris.

Figure 4 : Synthesis Dashboard of multimodal Accessibility in Paris

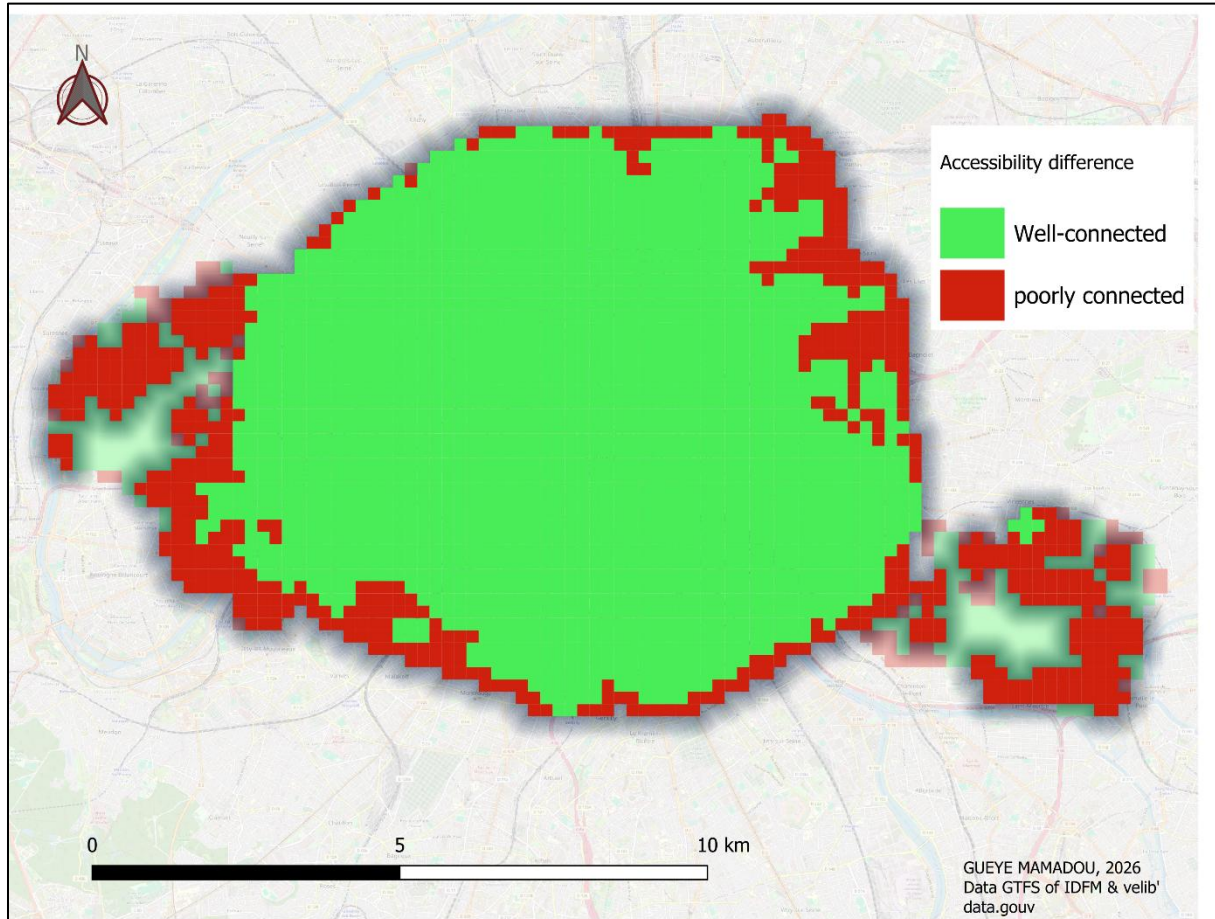


This six-panel synthesis dashboard provides a coherent and comparable overview. The top-left panel (scenario mean travel time) confirms the modal hierarchy: the full network (28.9 min) systematically outperforms partial configurations, and Vélib' alone (120 min) represents an impractical configuration at urban scale. The central panel (Gini index) reveals that inequality remains low in all scenarios including PT (0.069–0.081), while the Rail + Vélib' scenario is the most unequal among realistic scenarios.

The bottom-centre panel (travel time distribution for the reference scenario) reproduces the concentrated histogram already discussed, while the bottom-right panel (Lorenz curves) synthesises equity in a single comparative visualisation. The complete results table (bottom left) enables direct reading of all four scenarios across four dimensions:

mean time, Gini, % in desert, and gap versus the reference scenario. The 'gap to reference' column shows that Bus + Vélib' costs an extra 10.1 minutes compared to the full scenario, while TC no Vélib' costs only 1.6 minutes, confirming rail dominance.

3.3 Mobility Deserts / Identification and Location



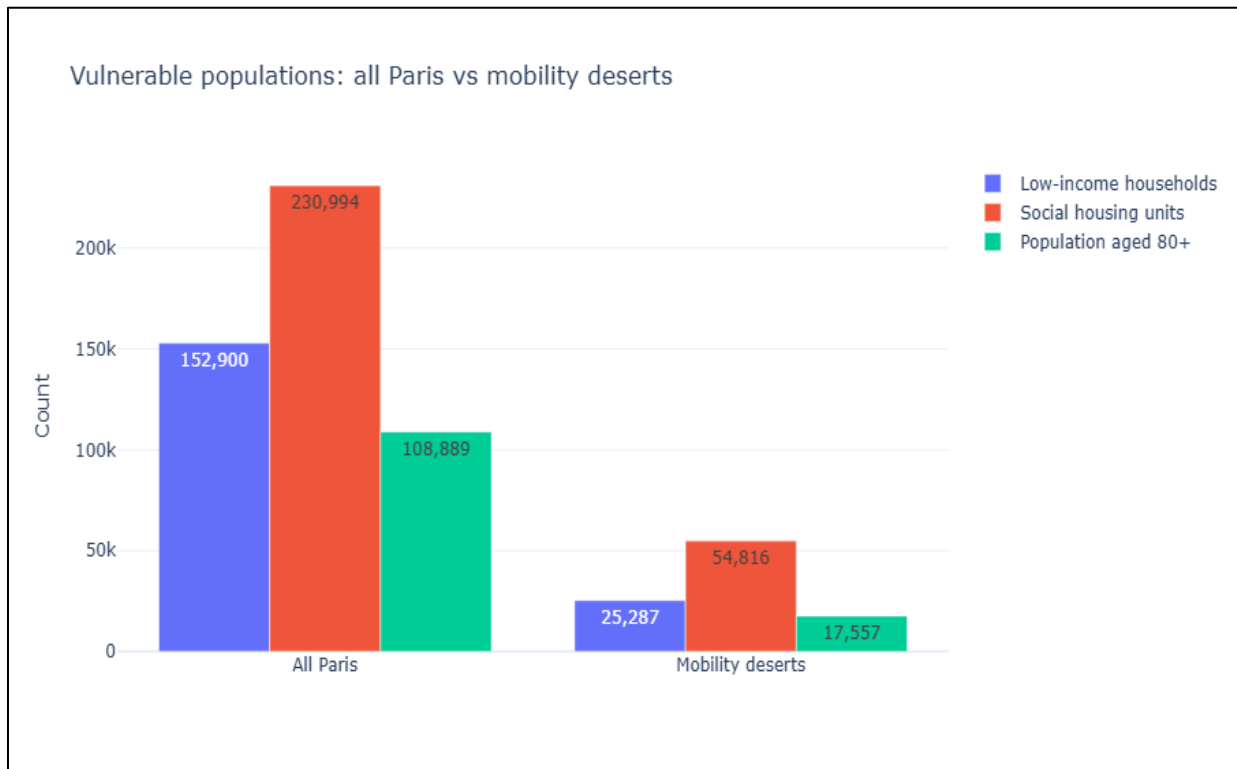
Map 8: location of mobility desert in Paris all transport mode combined

This binary map distinguishes well-connected zones (green) from poorly served areas or mobility deserts (red), defined as grid cells whose median travel time exceeds the 75th percentile of the distribution. The vast majority of inner Paris territory appears green, confirming the overall excellent level of the Parisian multimodal network. The centre-periphery gradient is here radically simplified into a dichotomous opposition that facilitates the identification of priority zones.

The mobility deserts (red) form a discontinuous ring along Paris's edges, particularly thick in the 16th arrondissement (Bois de Boulogne and around Porte de Saint-Cloud), in the north-east (Porte de la Chapelle sector), and in the 12th annex (Bois de Vincennes). These enclaves correspond to large green spaces, industrial zones, or areas undergoing

urban transformation that are poorly equipped with transport infrastructure. Their population, though a minority in volume, often cumulates other socio-economic vulnerability factors.

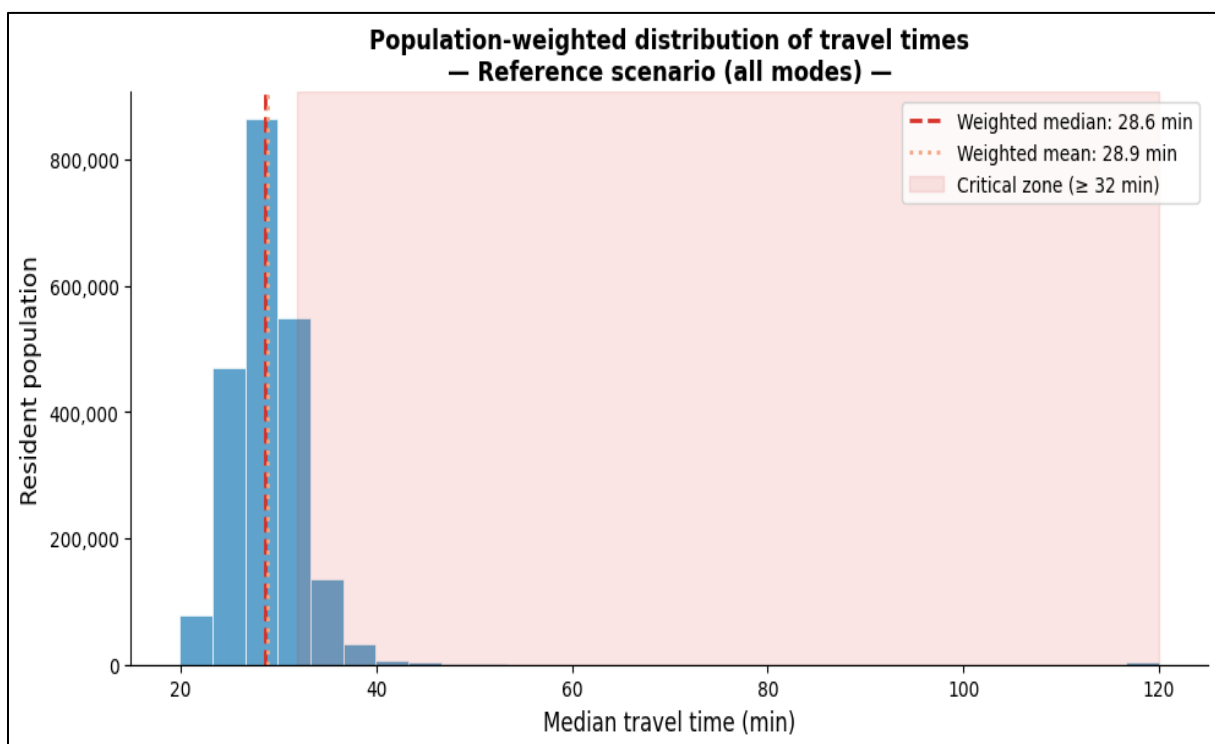
Figure 5: Vulnerable Populations: All Paris vs Mobility Deserts



This grouped bar chart compares the presence of vulnerable populations (low-income households, social housing units, people aged 80+) between the whole of Paris and mobility deserts only (cells above the 75th percentile of travel time). The slight over-representation of social housing in mobility deserts (23.7% versus 16.4% of total population in deserts) is a warning signal for public policy: social housing residents are statistically more exposed to accessibility deficits. However, the absence of strong over-representation of low-income households and elderly people in deserts suggests that social vulnerability and mobility vulnerability do not perfectly overlap in Paris, unlike what is observed in other metropolises. This moderate finding must nonetheless be interpreted cautiously, as mobility deserts include sparsely populated areas (green spaces) that dilute the figures.

Figure 6: Population-Weighted Distribution of Travel Times ; Reference Scenario (All Modes)

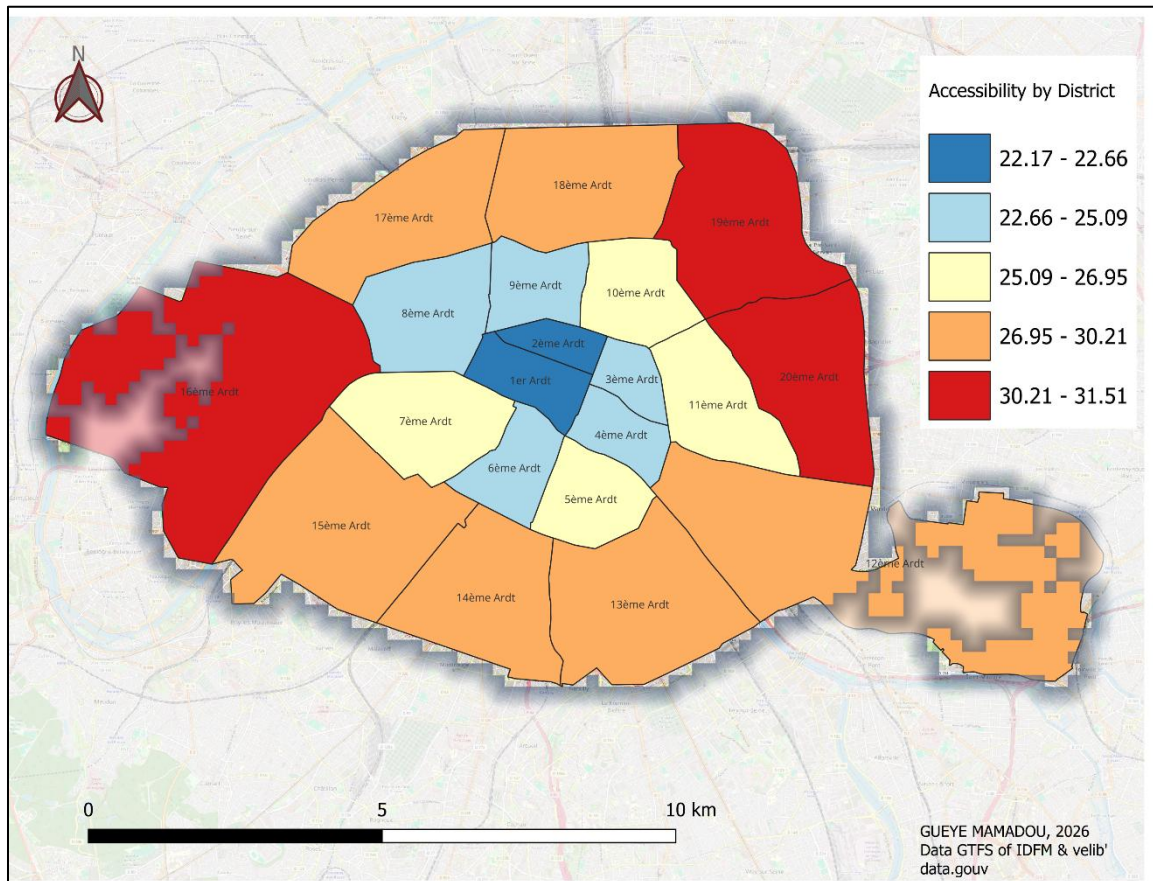
This population-weighted histogram shows how Parisian residents are distributed by their level of accessibility in the reference scenario. The distribution is strongly left-skewed and concentrated: the vast majority of residents (around 2 million) live in areas where median travel time sits between 25 and 32 minutes, with a peak at 28 minutes matching the weighted median. The weighted median (28.6 min) and weighted mean (28.9 min) are very close, indicating a near-symmetrical distribution for the main population.



3.4 Accessibility by Arrondissement; Population-Weighted Mean Travel Time

This choropleth map aggregates the 200m grid results at the scale of the 20 Parisian arrondissements by computing a population-weighted mean travel time. The central arrondissements (1st and 2nd, in dark blue, 22.2–22.7 minutes) benefit from the best multimodal accessibility, owing to the convergence of numerous metro lines and the maximum density of Vélib' stations. The second-ring arrondissements (5th–11th) display

intermediate times (22.7–26.9 minutes, light blue to pale yellow), reflecting a good service level but with greater internal disparities.



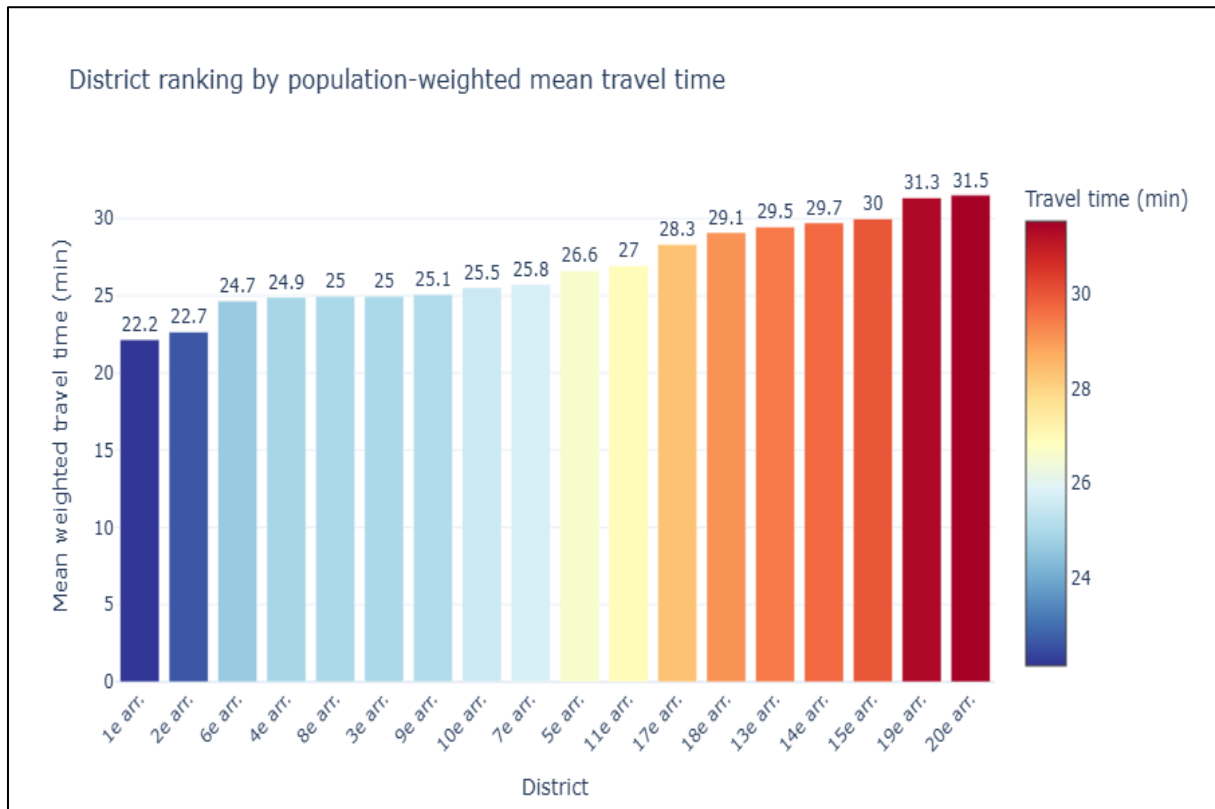
Map 9: Population-Weighted Mean Travel Time by arrondissement

The least well-served peripheral arrondissements are the 16th and 19th (30.2–31.5 minutes, bright red), followed by the 20th, 18th and 15th. This ranking reveals a persistent territorial inequality that cannot be explained solely by distance to the centre: the 19th and 20th, close to the inner suburbs, suffer from a relative deficit of direct metro lines, while the 16th is penalised by the presence of the Bois de Boulogne, which creates a spatial discontinuity in the network. This arrondissement-level map constitutes an effective communication and advocacy tool for directing mobility infrastructure investments.

Figure 7: District Ranking by Population-Weighted Mean Travel Time

This bar chart ranks all 20 Parisian arrondissements from most to least accessible by population-weighted mean travel time. The colour gradient (dark blue → bright red) immediately conveys the accessibility differential. The 1st arrondissement scores best

(22.2 min) and the 20th worst (31.5 min), a gap of 9.3 minutes between extremes. Central arrondissements (1st–4th, 6th, 8th) form a homogeneous high-accessibility cluster (22–25 min), while peripheral ones (16th, 19th, 20th) concentrate the worst scores.



The progression is not perfectly regular: a notable jump occurs between the 5th (25.8 min) and the 11th arrondissement (27 min), then a sharper acceleration from the 17th (28.3 min) onwards. This intermediate plateau corresponds to second-ring arrondissements that still benefit from decent metro coverage but begin to suffer from fine-grained service gaps. The 16th arrondissement (31.3 min), second-to-last, illustrates the negative effect of the Bois de Boulogne on network continuity.

Conclusion

This study mobilised a multimodal network model simulated on a 200m × 200m grid covering the entirety of inner Paris to address one central question: to what extent does the complementarity between soft mobility and public transit produce a differentiated accessibility across intra-Parisian space? The results converge on three major findings.

The rail network remains the irreplaceable backbone of Parisian accessibility. The reference scenario, combining all modes, yields a median travel time of 28.7 minutes and a Gini index of 0.070, the best performance across all dimensions. Removing the rail network (Bus + Vélib' scenario) degrades the median travel time to 39.2 minutes while simultaneously increasing spatial inequalities, with 92.8% of the population falling into the critical zone. Conversely, removing the bus alone (Rail + Vélib' scenario) maintains a median of 29.5 minutes, very close to the full scenario. This differential confirms that metro and tram are the structuring mode of Parisian accessibility, while the bus plays a role of fine-grained adjustment, useful but not decisive at city scale.

Vélib' fulfils a localised corrective function, not a redistributive one across Paris as a whole. Its overall contribution is modest, averaging a gain of 2.1 minutes for the entire population. However, its impact is remarkable in the two peripheral enclaves of the Bois de Boulogne (16th arrondissement) and the Bois de Vincennes (12th annex), where the combination of bike-share and bus reduces travel times by 50 to 88 minutes. In these mobility deserts, defined as cells exceeding the 75th percentile of the distribution, Vélib' represents the only means of substantially reducing travel times in the absence of a dense rail network. It therefore plays less a universal equity role than a territorial last-resort role in areas not covered by the metro. The Lorenz curves confirm this reading: Gini indices remain stable across scenarios that include PT, and the TC no Vélib' curve is almost perfectly superimposed on the reference curve. The paradox of the Vélib'-only scenario, with a Gini of 0 for a mean time of 120 minutes, illustrates with clarity that equality in deprivation is not equality in access.

Social vulnerability and mobility vulnerability do not systematically overlap in Paris, yet warning signals remain. The four scatter plots crossing social indicators with travel times display R^2 values close to zero (0.000 to 0.029), refuting the hypothesis of systematic accessibility discrimination against precarious populations. This result sets Paris apart

from many other metropolises where social geography and mobility geography mutually reinforce each other. However, the slight over-representation of social housing in mobility deserts (23.7% of the Parisian total versus 16.4% for the general population) is a signal that should not be dismissed. More significantly, the disaggregated analysis by social group reveals that low-income households lose an average of 15 additional minutes without the rail network, compared to 10.1 minutes for the overall population, indicating that they inhabit zones structurally more dependent on the metro. This heightened dependence exposes them more severely to service disruptions and line closures.

These findings open concrete public policy perspectives. First, the strong rail dependence of precarious populations argues for a policy of maintaining and prioritising metro development in the north-eastern arrondissements (18th, 19th, 20th), where the combination of social vulnerability and rail dependence is most pronounced. Second and finally, the limitations of this model deserve acknowledgement: the simulated travel times assume permanent and equal availability of all modes, without accounting for usage costs, physical barriers for elderly or mobility-impaired users, or temporal variations such as peak hours and night service. These qualitative and economic dimensions of accessibility constitute the primary avenue for deepening this research.

In sum, Paris presents a multimodal network that is globally performant and relatively equitable, but whose homogeneity conceals persistent peripheral inequalities and a strongly marked modal hierarchy. The complementarity between soft mobility and public transit does produce a differentiated accessibility across intra-Parisian space, not through general redistribution, but through localised correction at the margins of the urban territory where the rail network cannot extend.